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State of charge estimation for lithium-ion batteries based on improved barnacle mating optimizer and support vector machine

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ABSTRACT

Precise state of charge (SOC) estimation is essential for battery management systems. The improved barnacle mating optimizer-support vector machine (IBMO-SVM) model is proposed and used for SOC estimation of lithium-ion batteries. (1) The cubic chaotic mapping, hyperbolic sinusoidal conditioning factor, and Gauss-Cosey variation are introduced to improve the barnacle mating optimizer (BMO) to obtain the improved barnacle mating optimizer (IBMO); (2) The convergence performance of IBMO is verified by comparing with other five intelligent optimization algorithms under several test functions; (3) IBMO-SVM is created by using IBMO to optimize the search for support vector machine (SVM) parameters; and (4) IBMO-SVM is used for SOC estimation of lithium-ion batteries and the estimation results are analyzed by multiple evaluation indexes. The proposed model's root mean squared error (*RMSE*) and mean absolute percentage error (*MAPE*) are 0.0042 and 0.61 %, and its R-square (R^2) is 0.9994, outperforming the four comparison models. The SOC estimation methodology proposed in this study is highly accurate and reliable, and it provides advantages for improving the battery management system.

1. Introduction

As an energy storage device, lithium-ion battery has long service life, stable operating voltage, and green environment [1]. It is now widely utilized in electric power systems, electric vehicles [2], electronics [3], electric ship applications [4], and even in aerospace [5]. According to the "New Energy Vehicle Industry Development Plan (2021–2035)", new energy vehicle sales already account for 20 % of all new vehicle sales in 2022. By 2035, pure electric vehicles take up the majority of new vehicle sales [6]. Lithium-ion batteries have gradually replaced traditional energy sources as the main power source for automobiles.

In the literature, Chen et al. [7] argued that SOC is an important parameter to represent the remaining capacity of batteries, specifically, it is the ratio of the remaining capacity to the actual capacity of the battery. According to the SOC of the battery, Tan et al. [8] managed and adjusted the battery operation strategy more scientifically, which can enhance battery performance and lengthen battery life. With the commercialization of hybrid electric vehicles and plug-in hybrid electric vehicles in recent years, SOC estimation has become a critical issue in battery management systems. Inaccurate SOC estimations can lead to over-charging or over-discharging of lithium-ion batteries, which can irreparably harm them and significantly reduce their service life [9,10]. The accurate SOC estimation may alter the battery operating strategy in real time via the battery management system, allowing the lithium-ion battery to operate in the ideal state to reduce the system's energy loss while enhancing its safety and dependability. This has a good influence on supporting green development, energy savings, and emission reduction [11,12]. In practical applications, the battery exhibits a high degree of nonlinearity due to a series of factors such as current, voltage, charge/discharge multiplier, number of cycles, surface temperature, and battery life, which affects how accurately the battery's SOC is estimated. The objective of this study is to improve the accuracy of SOC estimation for lithium-ion batteries.

The methods for estimating SOC that are now in use broadly

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categorized into five groups in Fig. 1: direct discharge method, ampere-time integration method, voltage/impedance method, model estimation method, and machine learning method [13,14]. The direct discharge method, which obtains the battery's remaining capacity by discharging it, has a significant time lag and is hence typically only utilized for validation experiments [15]. The ampere-time integration method obtains the SOC by measuring the current and performing an integration calculation, but it is too dependent on the initial SOC value, and an inaccurate initial value might result in the accumulation of measurement errors later [16]. The voltage/impedance method obtains the SOC by calculating the voltage or impedance. The estimation results of this method are highly susceptible to the influence of the actual working environment of the battery, with large errors and lack of stability, so it is rarely applied. The model estimation method obtains the SOC in real time by associating the measured voltage, current, and temperature data with the mathematical model of the battery through the input measurement signal, and these methods mainly include: Kalman filter method and particle filter method [17,18]. This approach calculates the battery's current SOC in real time, regardless of the operating conditions, as long as a mathematical model that characterizes the battery operation and the measured battery signals are obtained. However, it relies too much on the mathematical model's accuracy and has high computational complexity. In addition, machine learning methods include fuzzy control, neural networks, and SVM. These methods analyze the characteristics of battery data through nonlinear mapping to approximate the dynamic characteristics of the battery. The advantage of machine learning methods is that the estimation accuracy is high and does not require explicit access to the battery's behavioral characteristics. Therefore, it is theoretically suitable for any chemistry type of battery [19]. The disadvantage is that it is more dependent on the quality of the training data and requires more stringent parameter settings for the machine model itself [20].

Most of the current SOC estimation methods published in journals by scholars are based on improvements of the above methods. For the Kalman filter method, Bi and Choe [21] proposed a Sigma point-based Kalman filter algorithm by approximating the distribution of the state variables of the actual nonlinear model, assigning corresponding weights to the sampling points, and updating the system state estimates by Kalman filtering after weighting. Kim and Cho [22] used the dual Kalman filter method to estimate the system state and parameters alternately through two independent Kalman filter algorithms, but it significantly increases the system's complexity and slows down the computational speed. Han et al. [23] proposed an adaptive Kalman filter to improve the estimation accuracy of the system by adaptively measuring and updating the noise covariance. These above-mentioned Kalman filter-based SOC estimation methods can perform more accurate estimation while the battery is operating, but they are still unable to get around the Kalman filter's reliance on an accurate battery mathematical model. The accuracy of the model determines the estimation accuracy of such methods to a certain extent, while they involve complex matrix calculations and therefore have high processor requirements. For machine learning models, Suykens et al. [24] proposed least-squares support vector machine (LSSVM) based on SVM to transform the inequality-constrained problem of SVM into an equation-constrained problem and the quadratic programming problem of SVM into solving a system of linear equations, thus improving the SOC estimation efficiency. Ali et al. [25] proposed a new method for adaptive SOC estimation based on Gaussian process regression, in which parameters such as voltage, capacity, and temperature of the battery are directly mapped into the corresponding model to achieve SOC estimation. Fan et al. [26] proposed a convolutional neural network-based SOC estimation method, which uses symmetric convolutional layers to solve the boundary effect, thus improving the accuracy of SOC estimation at the edges. However, the methods are based on the improvement of the machine model itself and do not address the fundamental problem of model dependence on the training data and the quality of the initial parameters, which will also lead to a complex tuning process when these methods are applied to SOC estimation.

With the inability of individual SOC estimation methods to address their inherent shortcomings, many researchers have focused on the hybrid estimation of multiple SOC estimation methods [27]. Misyris et al. [4] improved the accuracy of SOC estimation by using the ampere-time integration method for short-term SOC estimation and a linear Kalman filter for long-term SOC estimation to calibrate the SOC to eliminate cumulative errors. Wang et al. [28] corrected the ampere-time integration method with the open-circuit voltage method and effectively used to compensate for the accumulated error in the measurement when the accumulated settling time was out of the set range. In sum, these methods all use other SOC estimation methods to compensate for the accumulated error caused by the ampere-time integration method to improve the SOC estimation accuracy. However, they are more dependent on the measurement accuracy of the current sensor, while the

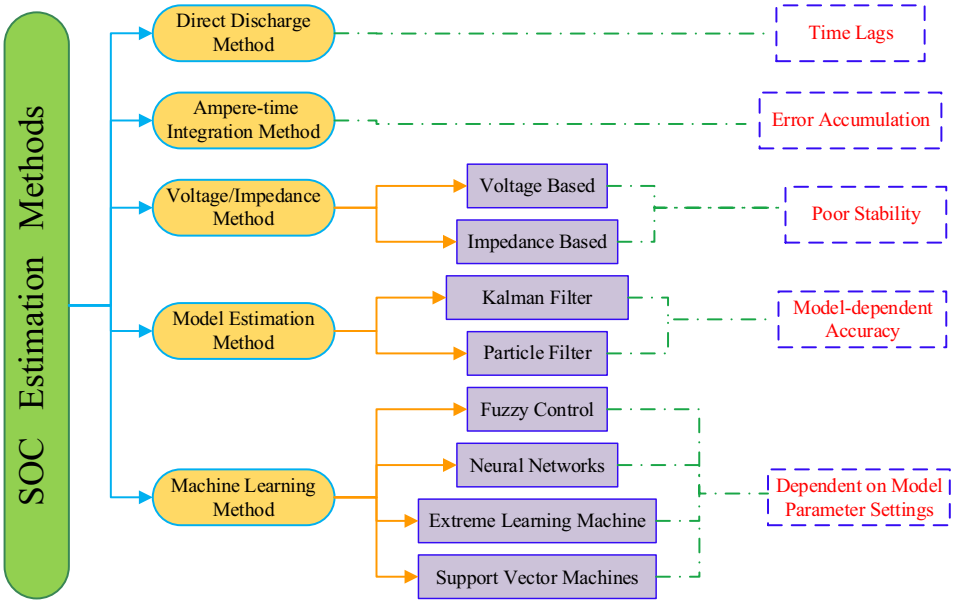


Fig. 1. Classification of SOC estimation methods.

current fluctuation caused by the unstable battery operating temperature will cause an increase in the measurement error. Lao et al. [29] and Xia et al. [30] combined variable forgetting factor recursive least squares and multi-forgetting factor recursive least squares algorithms with the traceless Kalman filter respectively thus enabling the SOC estimation accuracy to be improved, but amplifying the complexity of the traceless Kalman filter itself and making the operating time of the system significantly longer.

With the prevalence of intelligent optimization algorithms in recent years, many researchers have combined them with battery SOC estimation methods and achieved good results. Wei et al. [31] used the sine cosine algorithm for global optimization of dynamic neural network parameters, and by comparison, its SOC estimation outperformed the untraced Kalman filter. Li et al. [32] used particle swarm optimization (PSO) algorithm to optimize the kernel function of SVM to achieve improvement in the accuracy of SOC estimation, which effectively solved the parameter setting problem of SVM. Annamalai et al. [33] proposed a SOC estimation method based on stacked sparse autoencoder (SSAE) and salpine swarm algorithm (SSA). The parameters involved in SSAE are optimally tuned by using SSA to make a great improvement in SOC estimation performance. Maheshwari and Nageswari [34] combined the Sunflower Optimization (SFO) Algorithm with a machine learning model and thus performed SOC estimation, which outperformed the adaptive Kalman filter by the error metric analysis of the estimation results. All these methods combine intelligent optimization algorithms with machine learning models to solve the problem of machine model parameter setting, which effectively simplifies the tuning process of machine models and improves the accuracy and efficiency of SOC estimation. However, there is still room to improve the performance of these intelligent optimization algorithms, and there is still the problem of easily falling into local optimal solutions [51–53]. According to the no-free lunch theory of Service [35], no single algorithm can achieve good results in all optimization problems, and the algorithms show different optimization-seeking effects when dealing with different problems, so it is necessary to develop new optimization algorithms with stronger convergence performance when dealing with the lithium-ion battery SOC estimation problem. Sulaiman et al. [36] proposed BMO by studying the mating mechanism of barnacles, and the algorithm showed excellent merit-seeking ability in handling some test functions. In this study, IBMO is proposed by improving BMO and overcoming some of its convergence defects.

To improve the accuracy of SOC estimation for lithium-ion batteries, this study integrates the advantages and disadvantages of each SOC estimation method, and intends to solve the problem of machine model parameter setting by the optimization-seeking ability of the IBMO, to achieve better SOC estimation results. The objectives of this study are as follows.

- To propose strategies to improve BMO and thus enhance its convergence performance.
- To reduce the influence of random parameters on the SVM fitting results and improve its SOC estimation ability.
- To optimize the battery management system by improving the accuracy of SOC estimation to have a positive impact on energy storage devices.

The remaining sections of this study are organized as follows: Section 2 introduces the fundamental theory involved in this study; Section 3 introduces the process of proposing IBMO and verifies its performance; Section 4 shows the process of building the IBMO-SVM model and demonstrates its SOC estimation results through experiments; Section 5 summarizes the study and provides an outlook on the subsequent research.

2. Theoretical foundations

This section introduces the theoretical foundations of SVM as well as BMO.

2.1. SVM

SVM is a machine learning model built on VC dimensional theory and the principle of structural risk minimization. The core idea of it is to find an optimal classification hyperplane that maximizes the separation edge between the objects separated by the hyperplane [37]. SVM can avoid the "dimensional disaster" by mapping the nonlinear problem into a linear problem of higher dimensionality through the mapping function, so that the decision function is constructed in a higher dimension, and the idea is shown in Fig. 2.

The specific principle is that for a sample set $D = \{(x_i, y_i)|x_i \in R^d, y_i \in R\}$ ($i = 1, 2, 3, \ldots, N$), a mapping function $m(x)$ is established and $m(x)$ can transform the nonlinear problem into a linear problem at high dimensionality, resulting in the classification hyperplane shown below:

$$f(x) = \omega^T \cdot m(x) + \lambda \tag{1}$$

where ω is a weighting factor and λ is a constant.

The regression is transformed into a solvable optimization problem as follows:

$$\begin{cases} \min\limits_{\omega,\zeta_i} \dfrac{1}{2}\|\omega\|^2 + c\sum\limits_{i=1}^{N}\zeta_i \\ s.t. y_i\left(\omega^T \cdot m(x) + \lambda\right) \geq 1 - \zeta_i \end{cases} \tag{2}$$

where ζ_i is the relaxation factor and c is the penalty parameter.

Thereby its pairwise optimization problem is expressed as:

$$\begin{cases} \min\limits_{\alpha} \dfrac{1}{2}\sum\limits_{i=1}^{N}\sum\limits_{j=1}^{N}\alpha_i \cdot \alpha_j \cdot y_i \cdot y_j \cdot K\left(x_i, x_j\right) - \sum\limits_{i=1}^{N}\alpha_i \\ s.t. \sum\limits_{i=1}^{N}\left(\alpha_i \cdot y_i\right) = 0 \end{cases} \tag{3}$$

where $K(,)$ is the kernel function, which represents the inner product of feature vectors in the process of nonlinear transformation, thus effectively avoiding the complex operation in high-dimensional space and effectively solving the problem of "dimensional disaster", and α is the Lagrange multiplier.

Noble [38] concluded that the selection of the kernel function is important to SVM, and the most widely used kernel function is the radial basis function (RBF), whose expression is shown below:

$$K\left(x_i, x_j\right) = exp\left(-\gamma\left\|x_i - x_j\right\|^2\right) \tag{4}$$

where γ is the kernel width of RBF.

Penalty factor c and RBF kernel width γ is important to the generalization performance of SVM, and the reasonable selection of these two parameters can effectively improve the fitting performance of SVM. In this study, we intend to use these two parameters as optimization objects, to improve the estimation accuracy of SVM.

2.2. BMO

Barnacles are microorganisms that have existed since the Jurassic period [39]. The BMO is an intelligent algorithm inspired by the productive behavior of barnacles [36], which follows the following guidelines:

(1) The barnacle's mating group is restricted to all mates around its penis and any mate competitors that may exist around it.

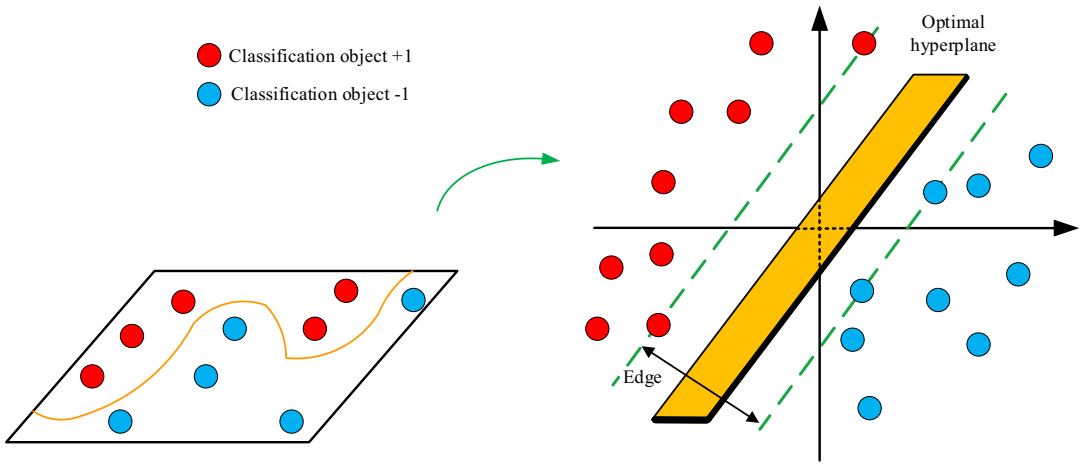


Fig. 2. Principle of SVM nonlinear mapping.

(2) The barnacle's mate selection process is random, but the mate selected must be restricted to the barnacle's penis length (*pl*).
(3) The barnacle may choose to offer or accept sperm during mating, but may accept sperm from other barnacles only once.
(4) Self-fertilization of barnacles is not considered.
(5) The sperm casting process occurs when all mate choices of the barnacle exceed *pl* during the iteration.

Based on the above criterion, the population vector of barnacles is described as:

$$\mathbf{X} = \begin{pmatrix} x_1^1 & \cdots & x_1^D \\ \vdots & \ddots & \vdots \\ x_d^1 & \cdots & x_d^D \end{pmatrix} \tag{5}$$

where D denotes the number of barnacle decision variables and d denotes the dimensionality of the barnacle. The upper and lower bounds of each decision variable obey the following equation:

$$ub = [ub_1, \ldots, ub_i] \tag{6}$$

$$lb = [lb_1, \ldots, lb_i] \tag{7}$$

where ub and lb represent the upper and lower bounds of the i-th decision variable.

The barnacle follows the following equation when selecting a mate:

$$b_f = rand(d) \tag{8}$$

$$b_m = rand(d) \tag{9}$$

where b_f denotes male barnacles, b_m denotes female barnacles, and d denotes the number of barnacles.

The process of barnacle population reproduction emphasizes the parental genetic characteristics, which is different from the replication process of other evolutionary algorithms. Based on the Hardy-Weinberg theorem, the expression for barnacle offspring generation is obtained as:

$$x_i^{D_{new}} = p x_{b_f}^D + q x_{b_m}^D \tag{10}$$

where p and q denote the proportion of male and female barnacle characteristics in newborn barnacles (p is a random number between 0 and 1, and $q = 1 - p$). ${x_{b_f}}^D$ denotes the barnacle parent parameter and ${x_{b_m}}^D$ denotes the barnacle female parameter.

Barnacle mate selection is restricted by the selection distance, and when the selection distance exceeds the preset *pl*, the offspring production is changed, which also represents the occurrence of the sperm casting process, with the following equation:

$$x_i^{d_{new}} = rand \times x_{b_m}^d \tag{11}$$

The Eq. (11) indicates that the new individuals produced during sperm casting are produced from female barnacles and inherit its characteristics. In each iteration of the algorithm, the optimal solution is placed at the top of $\boldsymbol{X}$, while the size of the newborn individuals is restricted and undesired individuals are sieved out.

3. IBMO and performance verification

This section describes the detailed steps to improve BMO and verifies the convergence performance of IBMO.

3.1. IBMO

Although BMO has strong performance in finding the optimum, its population diversity at the beginning of the iteration is insufficient, and its convergence speed is slow, while it is more likely to fall into local optimal solutions. The following improvements are made to address these shortcomings of BMO.

3.1.1. Cubic chaos mapping

Liu et al. [40] considered that the quality of the initialized population has a significant impact on the algorithm's search efficiency, and a uniform initial population can expand the range of the algorithm's search, thus effectively improving the algorithm's search speed. To address the shortcoming of insufficient population diversity at the beginning of BMO iteration, the cubic chaos mapping is introduced into the population initialization equation considering the characteristics of regularity and homogeneity of chaos operators. The cubic chaos mapping equation is expressed as follows:

$$t_{i+1} = 4{t_i}^3 - 3t_i \tag{12}$$

where, $-1 < t_i < 1$ and $t_i \neq 0$.

The population initialization process based on cubic chaos mapping is as follows: first a d-dimensional vector of individuals with absolute values not exceeding 1 is generated, then the remaining D-1 dimensional individuals are obtained by applying Eq. (12) to iterate over the individuals in each dimension, and finally the chaotic variables are mapped onto the barnacle individuals by applying the following equation:

$$\mathbf{X} = lb + (ub - lb) \times (t + 1)/2 \tag{13}$$

where $\boldsymbol{X}$ is the mapped population and t is the cubic chaos mapping factor.

3.1.2. Hyperbolic sine control factor

Taking advantage of the oscillatory nature of the hyperbolic sine function, the hyperbolic sine modulation factor optimization strategy is introduced in this study [41], which is equated as follows:

$$ss_l = 1 - \frac{Ite_max}{5} \times sinh\left(\frac{l^3}{100 \cdot Ite_max^3}\right) \tag{14}$$

where l is the current number of iterations of the algorithm and Ite_max is the maximum number of iterations.

Applying Eq. (14) to the offspring generation equation for BMO (Eq. (10)) yields the following expression:

$$x_i^{D_{new}} = ss_l x_{b_f}^D + (1 - ss_l) x_{b_m}^D \tag{15}$$

At the beginning of the iteration, the value of ss_l is large, and the algorithm has a large search step, thus improving the search range; at the end of the iteration, the decreasing speed of ss_l becomes faster, thus the convergence of the algorithm is accelerated and the efficiency of the algorithm is improved.

3.1.3. Gauss-Cosey variation

To address the drawback that BMO is easy to fall into local optimum, Gaussian variance and Cosey variance are introduced in this study. When the optimal solution of the algorithm falls into stagnation, the Gaussian variation is preferentially used for it. The Gaussian variation operator enables the algorithm to focus its search on the local area near the barnacle individuals, thus enhancing the local search ability of the algorithm [42]. The Gaussian operator obeys the following probability density equation:

$$G(v) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(-\frac{(v-u)^2}{2\sigma^2}\right) \tag{16}$$

where σ is the standard deviation and u is the mathematical expectation.

The Gaussian operator is used for population update when the barnacle optimal solution stalls, and the equation is expressed as follows:

$$x_i^{D_{new}'} = x_i^{D_{new}} + \lambda^* G(0,1) \tag{17}$$

where $G(0,1)$ obeys the Gaussian distribution and λ is a constant.

When the algorithm optimal solution update stalls again after Gaussian variation, the Cosey variation is used for it. The Cosey variation operator has a longer step length, which gives it a stronger perturbation capability and thus allows the algorithm to have a higher probability of jumping out of the local optimal solution [43]. The Cosey operator obeys the following probability density equation:

$$K(v) = \frac{1}{\pi \cdot (1 + v^2)} \tag{18}$$

Applying it to population renewal, the equation is expressed as follows:

$$x_i^{D_{new}''} = x_i^{D_{new}'} + \varphi^* K(0,1) \tag{19}$$

where $K(0,1)$ obeys the Cosey distribution and φ is a constant.

3.1.4. IBMO operation process

The specific running process of IBMO is shown below:

(1) Initialize the algorithm parameters and set the parameters such as the maximum number of iterations, the upper and lower limits, the number of populations, and *pl*.
(2) Generating initial populations using the cubic chaos mapping method, after which the initial fitness values of the barnacles are calculated.
(3) Sorting the fitness values of the barnacles and placing the barnacle with the best fitness value at the top of the population $\boldsymbol{X}$.
(4) Selecting barnacle females and males for mating.
(5) Determine whether the distance between female and male barnacles exceeds the preset value *pl*.
(6) When the distance between female and male barnacles is less than *pl*, update the offspring of barnacles using hyperbolic sine conditioning factor.
(7) When the distance between male and female barnacle homes is greater than *pl*, the sperm casting process occurs to renew the offspring.
(8) Calculate the offspring fitness value and update $\boldsymbol{X}$.
(9) Determine whether the iteration is stalled, and when the iteration is stalled, generate a new population using Gauss-Cosey variation, calculate the new population fitness value and compare it with the fitness value of the original population, retain the better population, and update $\boldsymbol{X}$.
(10) Determine whether the maximum number of iterations is reached. If the maximum number of iterations is reached, the algorithm is terminated and the optimal solution is output; otherwise, return to step (4) and continue iteration.

The specific operation of IBMO is shown in Fig. 3.

3.2. IBMO performance verification

Six standard test functions are selected to verify the convergence performance of IBMO, among which, f_1, f_2, and f_3 are single-peak test functions, and f_4, f_5, and f_6 are multi-peak test functions. The basic information of each test function is shown in Table 1. In this study, PSO [44], SFO [45], GWO [46], IGWO [47], and BMO were selected as comparisons for IBMO convergence performance for further validation. All algorithms were tested using Matlab-R2017a programming, computer operating system was Windows 10, processor was Inter Core i5-6300HQ 12GB, number of algorithm iterations were set to 500, population was set to 30, other parameters were set as shown in Table 2. The settings and definitions of the parameters in Table 2 are taken from the original literature of each algorithm, in which the convergence performance of each algorithm has been confirmed, thus ensuring the fairness of the algorithm comparison experiments in this study.

Each algorithm was run 50 times independently for the test functions, and the optimal value, average value, standard deviation, and average time(s) were recorded, and the test results are shown in Table 3.

From Table 3, the optimal values, standard deviations, and averages of IBMO in solving the single-peak test functions f_1, f_2, and f_3 are optimal among all tested algorithms, which proves that its local search capability is better than other comparative algorithms in the current test environment. In solving the multi-peak test functions f_4, f_5, and f_6, IBMO has the optimal standard deviation and average value compared with other algorithms, which proves that it has stronger global search capability in the current test environment and can effectively avoid some local optimal solutions. Meanwhile, the average running time can reflect the operational complexity of the algorithms. From Table 3, the average running time of IBMO is still in the same order of magnitude relative to other algorithms, indicating that the improvement strategy proposed in this study enhances the convergence performance of IBMO without making a significant increase in its operational complexity.

To show the convergence performance of IBMO more intuitively, Fig. 4 lists the convergence curves of each algorithm when solving the test function. The figure shows that whether solving single-peak test functions f_1, f_2, and f_3, or multi-peak test functions f_4, f_5, and f_6, IBMO has higher convergence accuracy and faster convergence speed, and can complete convergence faster and more accurately.

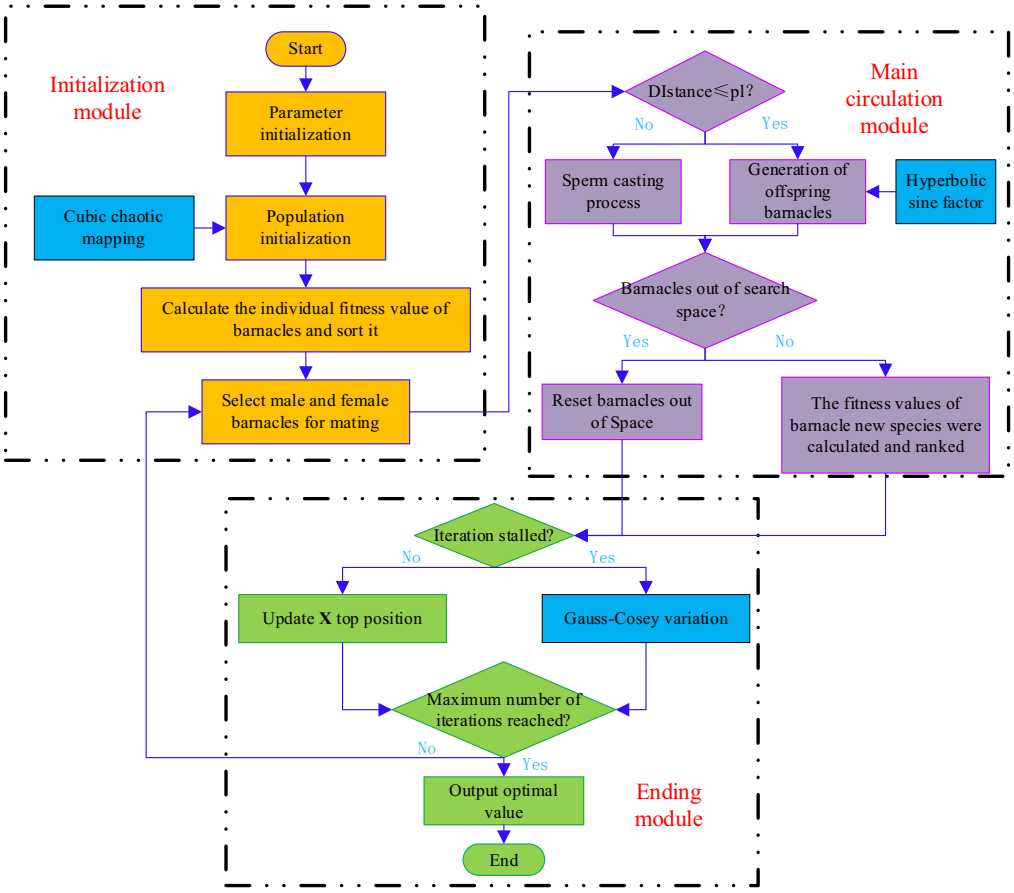


Fig. 3. IBMO operation flow chart.

Table 1
Test function information.

Function type	Function expression	Optimal solution	Lower limit	Upper limit
Unimodal	$f_1 = \sum_{i=1}^{\dim} v_i^2$	0	−100	100
	$f_2 = \max_i\{\lvert v_i\rvert, 1 \le i \le \dim\}$	0	−100	100
	$f_3 = \sum_{i=1}^{\dim} iv_i^4 + rand\ [0,1)$	0	−1.28	1.28
Multimodal	$f_4 = \sum_{i=1}^{\dim} \frac{v_i^2}{4000} - \prod_{i=1}^{\dim} cos\left(\frac{v_i}{\sqrt{i}}\right) + 1$	0	−600	600
	$f_5 = \sum_{i=1}^{\dim}(v_i^2 - 10 * \cos(2\pi v_i) + 10\dim)$	0	−5.12	5.12
	$f_6 = -20*exp\left(-0.2*\sqrt{\frac{1}{dim}\sum_{i=1}^{\dim} v_i^2}\right) - exp\left(\frac{1}{dim}\sum_{i=1}^{\dim} cos(2\pi v_i)\right) + e + 20$	0	−32	32

Table 2
Parameter settings of each algorithm.

Algorithm	Parameters	Algorithm	Parameters
PSO	$c_1 = c_2 = 1.48$, $w_{max} = 0.9$, $w_{min} = 0.1$	IGWO	$A = 2a * rand(0,1) - a$, $C = 2 * rand(0,1)$
SFO	$A = 4$, $Sp = 0.3$	BMO	$pl = 7$
GWO	$A = 2a * rand(0,1) - a$, $C = 2 * rand(0,1)$	IBMO	$pl = 7$, $\lambda = 1.5$, $\varphi = 1.5$

4. Experimental testing and evaluation

This section describes the process of acquiring lithium-ion battery data, the process of building lithium-ion battery SOC estimation model, and the analysis and evaluation of the model estimation results.

4.1. Acquisition of lithium-ion battery operating conditions data

In this study, the ternary lithium-ion battery of 18,650 type is taken as an example, and its main performance parameters are shown in Table 4. The current, voltage and temperature data of the battery are measured by NEWARE BTS-4008 battery test system, its appearance is shown in Fig. 5. Since the SOC data of each sampling point cannot be measured directly, the current SOC of the battery is obtained by the ampere-time integration method, and its equation is shown below:

$$SOC_{k+1} = SOC_k + \frac{\Delta t \cdot i_L}{C_N} \tag{20}$$

where k is the kth sampling with Δt as the sampling interval, i_L is the current during charging and discharging, when i_L is >0, the battery is charged, and when i_L is <0, the battery is discharged. C_N is the rated

Table 3
Algorithm test results.

Function	Statistical values	PSO	SFO	GWO	IGWO	BMO	IBMO
f_1	Optimum value	3.31E-06	1.35E-13	5.39E-30	8.72E-96	0.00E+00	0.00E+00
	Average value	1.20E-04	6.69E-11	1.57E-27	6.41E-92	0.00E+00	0.00E+00
	Standard deviation	1.36E-04	7.73E-11	2.89E-27	3.36E-91	0.00E+00	0.00E+00
	Average time(s)	0.382	0.563	0.442	0.551	0.458	0.482
f_2	Optimum value	7.67E-01	8.10E-08	6.94E-08	1.15E-37	1.57E-306	0.00E+00
	Average value	1.10E+00	1.33E-06	7.72E-07	5.79E-35	1.48E-285	0.00E+00
	Standard deviation	2.08E-01	1.43E-06	1.16E-06	1.12E-34	0.00E+00	0.00E+00
	Average time(s)	0.325	0.464	0.396	0.464	0.417	0.424
f_3	Optimum value	6.86E-02	1.12E-05	6.74E-04	1.03E-05	1.77E-07	1.65E-08
	Average value	1.77E-01	5.82E-04	1.94E-03	3.35E-04	3.71E-05	3.01E-05
	Standard deviation	6.11E-02	4.65E-04	7.54E-04	2.72E-04	3.43E-05	2.25E-05
	Average time(s)	0.335	0.526	0.417	0.491	0.473	0.454
f_4	Optimum value	1.31E-06	2.09E-15	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	Average value	5.77E-03	3.47E-12	6.49E-03	1.51E-03	0.00E+00	0.00E+00
	Standard deviation	1.02E-02	5.69E-12	1.20E-02	4.77E-03	0.00E+00	0.00E+00
	Average time(s)	0.338	0.434	0.373	0.437	0.391	0.404
f_5	Optimum value	2.43E+01	2.58E-11	5.68E-14	0.00E+00	0.00E+00	0.00E+00
	Average value	5.83E+01	4.80E-07	3.43E+00	0.00E+00	0.00E+00	0.00E+00
	Standard deviation	1.76E+01	8.38E-07	4.46E+00	0.00E+00	0.00E+00	0.00E+00
	Average time(s)	0.336	0.405	0.386	0.414	0.382	0.377
f_6	Optimum value	4.04E-03	1.98E-07	7.55E-14	4.44E-15	8.88E-16	8.88E-16
	Average value	2.74E-01	3.41E-06	1.02E-13	6.10E-15	7.36E-16	8.88E-16
	Standard deviation	5.11E-01	2.93E-06	1.65E-14	1.80E-15	4.01E-31	1.00E-31
	Average time(s)	0.371	0.554	0.507	0.623	0.548	0.562

capacity of battery. The battery used in this study has a long cycle life (>1500 times), so it is assumed that the actual capacity of the battery remains constant during the experimental measurements [4]. Therefore, the capacity of the battery in the equation takes the rated capacity.

The charging and discharging experiments were conducted, and the experimental measured current, voltage, temperature, and other working condition data were recorded based on the battery resting for 2 h. The SOC data of the battery was calculated by the ampere-time integration method to approximate the actual SOC of the battery. A set of working condition battery data was measured as shown in Fig. 6. The average ambient temperature during the entire test was 24 °C. In the Fig. 6, (a) is the working condition current data, (b) is the working condition voltage data, (c) is the working condition temperature data, and (d) is the SOC data calculated by the ampere-time integration method. The working condition data in the figure are collected with a sampling interval of 1 s and the initial value of SOC is 1. A total of 32,851 data were obtained, which is sufficient to support the training of the machine learning model and can effectively reduce the chance of the experiment.

4.2. SOC estimation modeling

Based on the working condition data of 18,650 ternary lithium battery, an IBMO-SVM model is proposed to estimate its SOC in this study. Firstly, the working condition data are divided into training samples and test samples; secondly, the training samples are input into the SVM for fitting training, and the optimal SVM parameters are found by iterating through the IBMO; finally, the current, voltage, and temperature of the test samples are input as input variables to SVM with optimal parameters, and then output the estimated SOC of the test sample.

Based on the above idea, the *RMSE* value of SOC is chosen as the *fitness* value of IBMO-SVM, and its equation is shown below:

$$fitness = \sqrt{\frac{1}{A}\sum_{k=1}^{A}\left(SOC_p{}^{(k)} - SOC_r{}^{(k)}\right)^2} \tag{21}$$

where A denotes the number of samples, $SOC_r{}^{(k)}$ denotes the actual SOC at the kth sampling point, and $SOC_p{}^{(k)}$ denotes the model output SOC at the kth sampling point.

The specific process of SOC estimation based on IBMO-SVM is as follows:

(1) The battery working condition data are divided into training samples and test samples. To improve the robustness of the algorithm estimation, the working condition data are randomly disordered, and the first 50 % of the disordered data are used as training samples and the second 50 % as test samples, and the order before disordering is restored after the estimation is completed.
(2) Normalize the data and remove their magnitudes.
(3) Initialize the parameters of SVM and IBMO.
(4) Input the training sample data into the SVM for training, and iterate through the IBMO to find the optimal SVM parameters (penalty factor c and RBF kernel width γ).
(5) The parameters obtained from the IBMO search are input to the SVM and the model is tested with test samples.
(6) Output the estimated SOC of the model and end the program.

Fig. 7 shows the flow chart of IBMO-SVM.

4.3. Experimental results and analysis

IBMO-SVM is used to estimate the SOC for the battery conditions shown in Fig. 6. SVM, Back Propagation (BP) neural network [48], PSO-SVM, and BMO-SVM are used as comparison models to verify the estimation performance of IBMO-SVM. The population number of each algorithm is set to 20 and the number of iterations is set to 100. Other parameters of each model were set as shown in Table 5. The settings and definitions of the parameters in Table 5 are taken from the original literature of each algorithm, and the same upper and lower limits of parameter search are set for the SVM to ensure the fairness of the comparison experiments.

The lithium-ion battery SOC estimation results of the five models with set parameters are shown in Fig. 8. In the figure, the SVM model (orange solid line) has the most divergent estimation curve and the estimation effect is poor to meet the practical needs, while the SOC estimation results of the IBMO-SVM model (blue solid line) and the BMO-SVM (red solid line) are closest to the actual values (black dotted line) and have better estimation effects compared with the other three models. Compared with the BMO-SVM model, the estimation effect of the IBMO-SVM model is further improved, thanks to the better global

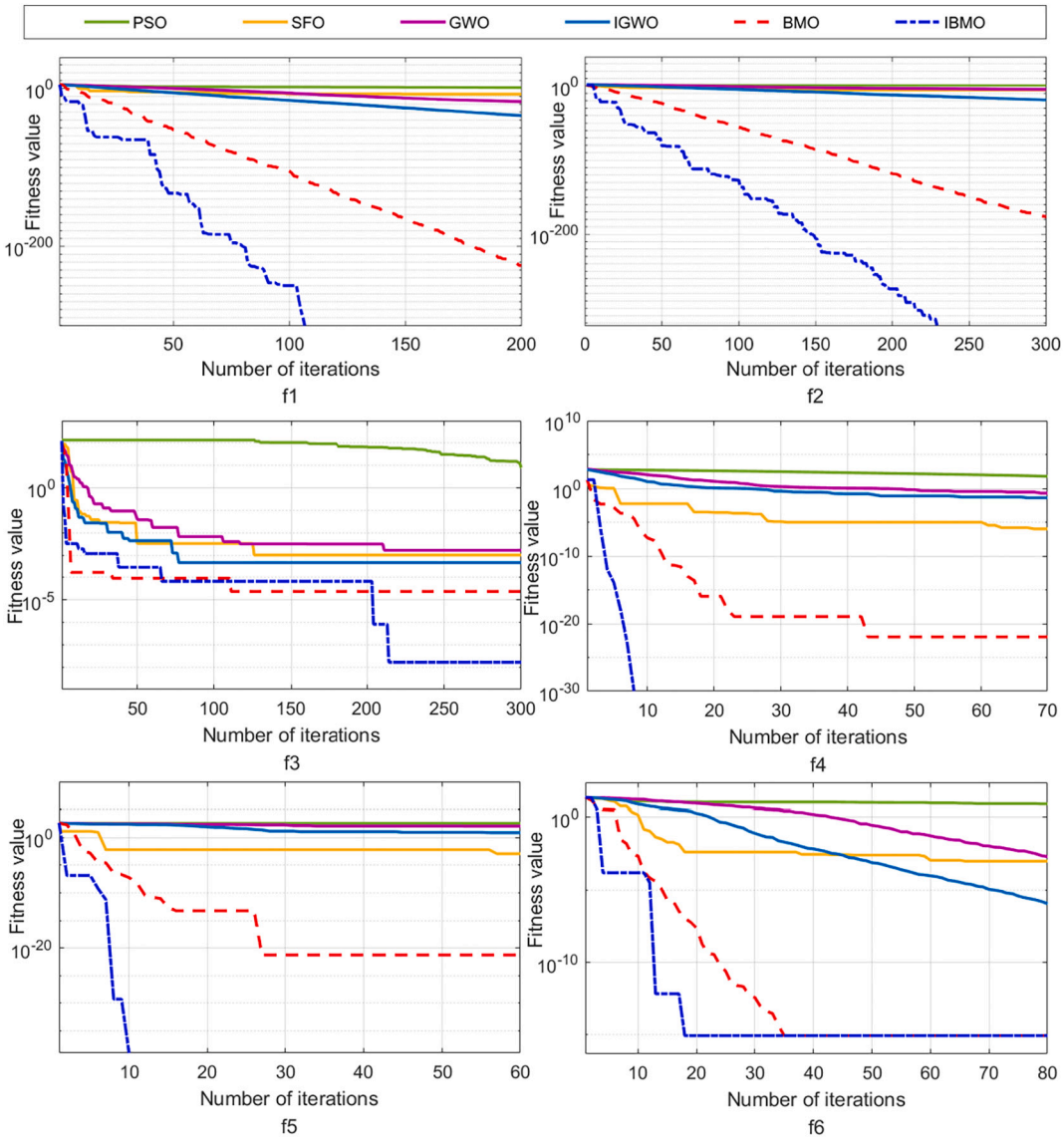


Fig. 4. Comparison of algorithm convergence curves.

Table 4
18,650 ternary lithium-ion battery main performance parameters.

Battery parameter	Value	Battery parameter	Value
Rated capacity	2000 mAh	Rated voltage	3.7 V
Charge cut-off voltage	4.2 V	Discharge cut-off voltage	2.5 V
Charge cut-off current	51 mA	Weight	46 ± 2 g

search capability of IBMO compared with BMO.

Meanwhile, Fig. 9 shows the errors between the estimation results of the five models and the actual SOC. The figure shows that the error curve of the IBMO-SVM model (blue solid line) is closest to 0 with the least fluctuation, and the estimation results of the SVM and BP models have large fluctuations. The main reasons for the fluctuation of model estimation results are: (1) The volatility of the voltage and current itself under the actual working conditions. (2) The selection of estimation model parameters has a large influence on the actual estimation results. The SVM and BP models used for experiments do not have the ability to select suitable parameters, so the estimation results have a large randomness, while the PSO-SVM, BMO-SVM, and IBMO-SVM choose to select the parameters with intelligent algorithms, so the estimation



Fig. 5. Appearance of NEWARE BTS-4008.

results are more stable, which proves that it is important to optimize the parameters of traditional machine learning models.

Fig. 10 and Table 6 show the estimation error distribution for each SOC estimation model, where *PE* denotes the percentage error of the estimation result, and the equation is shown below.

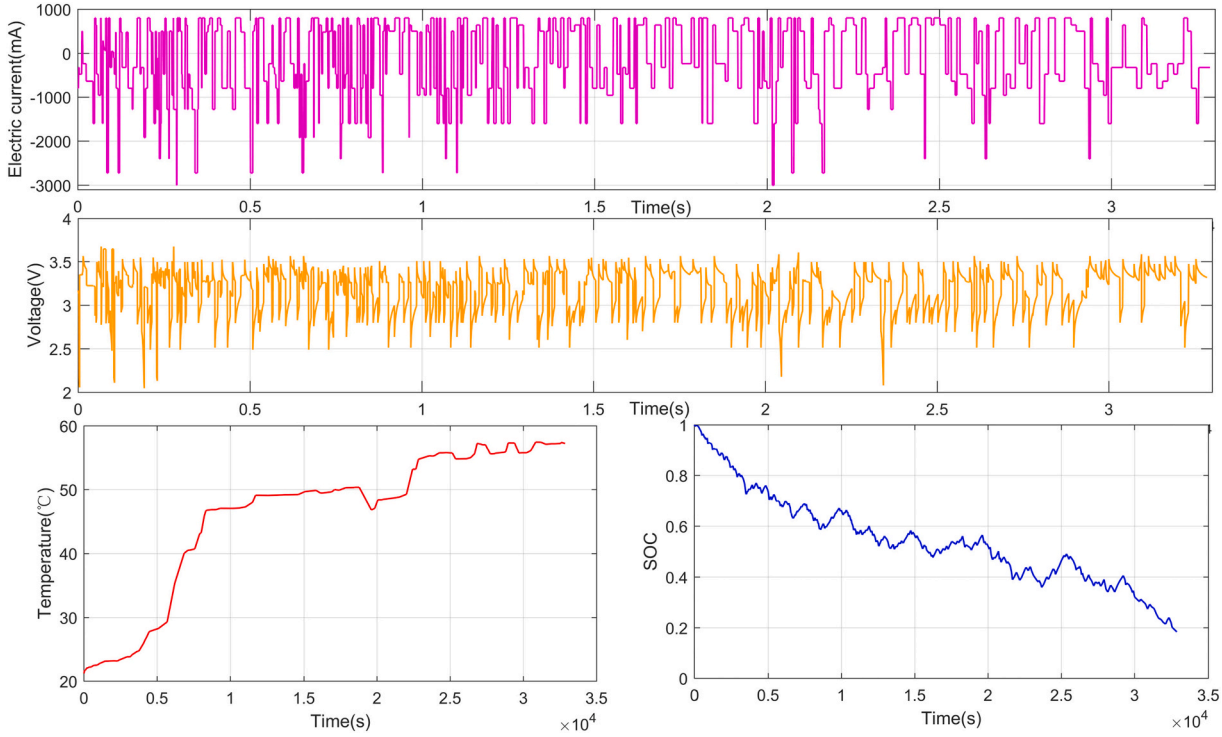


Fig. 6. Battery operating conditions data.

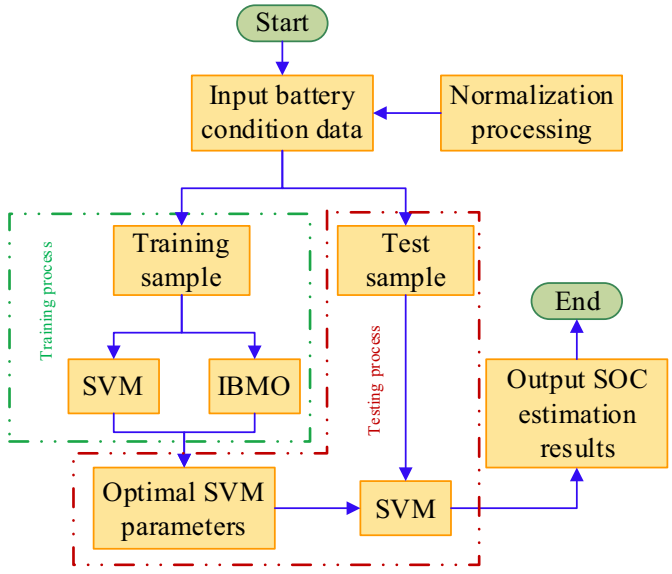


Fig. 7. Flow chart of IBMO-SVM.

Table 5
Model parameters settings.

Model	Parameters
SVM	$c = 1, \gamma = 0.01$
BP	$Ir = 0.05, s = 50, ep = 1000$
PSO-SVM	$c_1 = c_2 = 1.48, w_{max} = 0.9, w_{min} = 0.1, c_{min} = 0.1,$ $c_{max} = 100, \gamma_{min} = 0.01, \gamma_{max} = 1000$
BMO-SVM	$pl = 7, c_{min} = 0.1,$ $c_{max} = 100, \gamma_{min} = 0.01, \gamma_{max} = 1000$
IBMO-SVM	$pl = 7, \lambda = 1.5, \varphi = 1.5, c_{min} = 0.1,$ $c_{max} = 100, \gamma_{min} = 0.01, \gamma_{max} = 1000$

$$PE^{(k)} = \left|\frac{SOC_p{}^{(k)} - SOC_r{}^{(k)}}{SOC_r{}^{(k)}}\right| * 100\% \tag{22}$$

where $SOC_r{}^{(k)}$ denotes the actual SOC at the kth sampling point, and $SOC_p{}^{(k)}$ denotes the model output SOC at the kth sampling point, $PE^{(k)}$ denotes the PE value of the kth estimated result.

From Table 6, among the 16,425 estimation samples, IBMO-SVM has 9480 estimation results with $PE < 0.5$ %, which is better than other comparison models, while the number of its estimation results with $PE > 2$ % is only 605, which is much less than other comparison models. From Fig. 10, the PEs of IBMO-SVM estimation results are concentrated in the part of <1 %, accounting for 80 % of the total estimated samples, which proves the accuracy and stability of its SOC estimation.

To further demonstrate the estimation results of each model, the $RMSE$, R^2, $MAPE$, cumulative error (CE), and Running time are selected to evaluate the estimation results of the models. Among them, R^2 is used to assess the degree of fit between the curves, $RMSE$ and $MAPE$ is used to assess the estimation accuracy of the model, and CE is used to assess the estimation stability of the model. The expressions of each error evaluation index are as follows:

$$RMSE = \sqrt{\frac{1}{A}\sum_{k=1}^{A}\left(SOC_p{}^{(k)} - SOC_r{}^{(k)}\right)^2} \tag{23}$$

$$R^2 = 1 - \frac{\sum_{k=1}^{A}\left(SOC_p{}^{(k)} - SOC_r{}^{(k)}\right)^2}{\sum_{k=1}^{A}\left(SOC_p{}^{(k)} - \overline{SOC_r{}^{(k)}}\right)^2} \tag{24}$$

$$MAPE = \frac{100\%}{A}\cdot\sum_{k=1}^{A}\left|\frac{SOC_p{}^{(k)} - SOC_r{}^{(k)}}{SOC_r{}^{(k)}}\right| \tag{25}$$

$$CE = \sum_{k=1}^{A}\left|SOC_p{}^{(k)} - SOC_r{}^{(k)}\right| \tag{26}$$

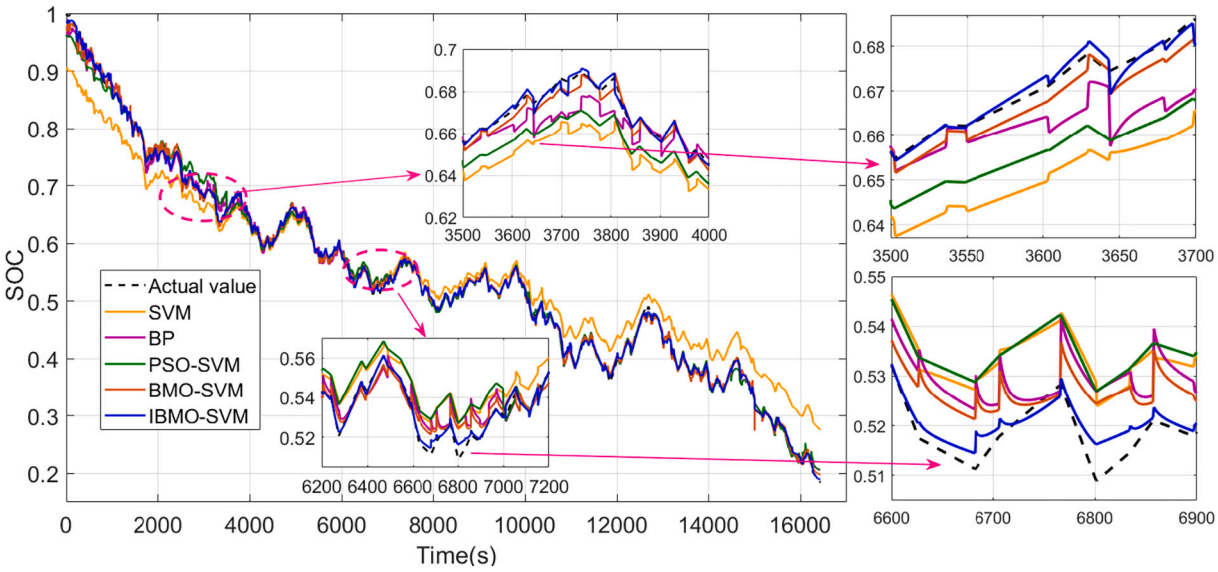


Fig. 8. SOC estimation results for each model. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

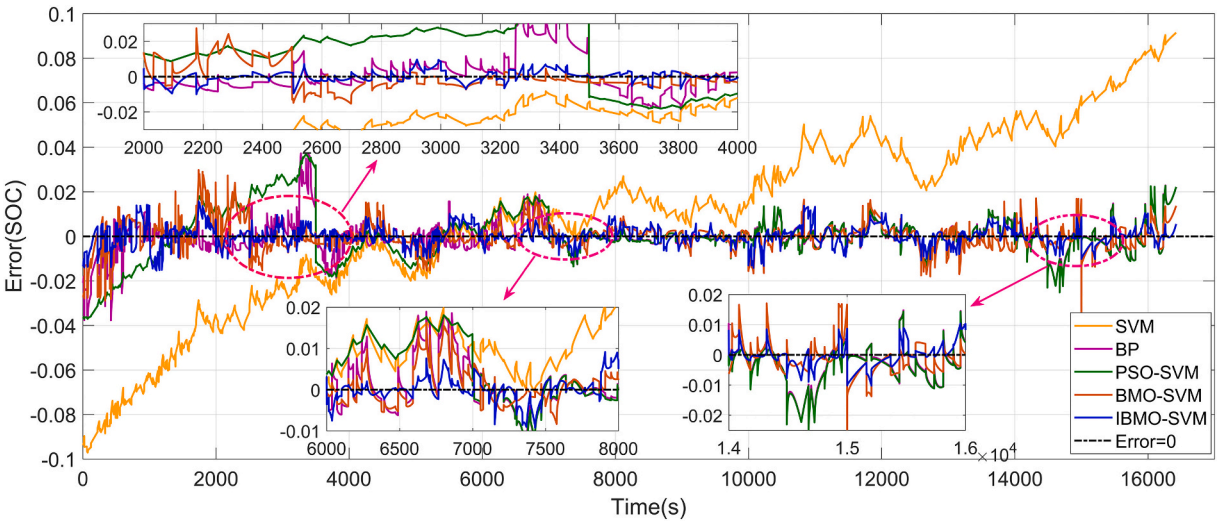


Fig. 9. SOC estimation error for each model. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

where A denotes the number of samples, and $\overline{SOC_r^{(k)}}$ is the mean value of $SOC_r^{(k)}$.

Table 7 shows the assessment results of each error evaluation index. The table intuitively shows that the *RMSE* and *MAPE* of the IBMO-SVM model are only 0.0042 and 0.61 %, which are the smallest among the five estimated models, proving that the estimated results of the IBMO-SVM model have the smallest deviation from the true values. Meanwhile, the R^2 of IBMO-SVM model is 0.9994, which is the closest to 1 among the five models, proving that it has the best fitting effect and the estimated curve is the closest to the actual value curve. The *CE* of IBMO-SVM model is only 50.7099, which is also the best among the five models, proving that its estimation results are the most stable and the overall estimation is less volatile than other estimation models. Meanwhile, compared with PSO-SVM and BMO-SVM, which also require algorithmic iterative optimization search, the running time of IBMO-SVM does not improve significantly, indicating that the time complexity of these three models is still in the same order of magnitude.

In summary, the IBMO-SVM model has better SOC estimation, with *RMSE* reduced by 0.0367, 0.0103, 0.003, 0.0021, and *CE* reduced by 494.3431, 80.3974, 28.1324, 21.1082, compared with SVM, BP, PSO-SVM, and BMO-SVM. The estimation accuracy and stability have been significantly improved compared with other models.

In addition, the SOC estimation method proposed in this study is still competitive when compared with similar studies in published papers by other researchers. He et al. [49] combined the extended Kalman algorithm and the Sage-Husa filter algorithm to develop a SOC estimation model for lithium-ion batteries, and the *PE* of its 83 % SOC estimation result is controlled within 1.5 % and by Fig. 10, the proposed IBMO-SVM model in this study is 91 %. Ling and Wei [11] used a double extended Kalman filter to estimate the lithium-ion battery SOC, and the *RMSE* of its estimation result is 0.0063, while the *RMSE* of the IBMO-SVM estimation result in this study is 0.0042. Li et al. [50] used a mixture of algorithms to estimate the SOC, and the *RMSE* of the estimation results was 0.009. Ali et al. [25] used an adaptive SOC estimation based on Gaussian process regression, and the *MAPE* of the estimation results was 2 %. In summary, compared with the SOC estimation methods proposed by other researchers in published papers, the method proposed in this study is still competitive enough in terms of the values of each error evaluation index to prove its accuracy and reliability.

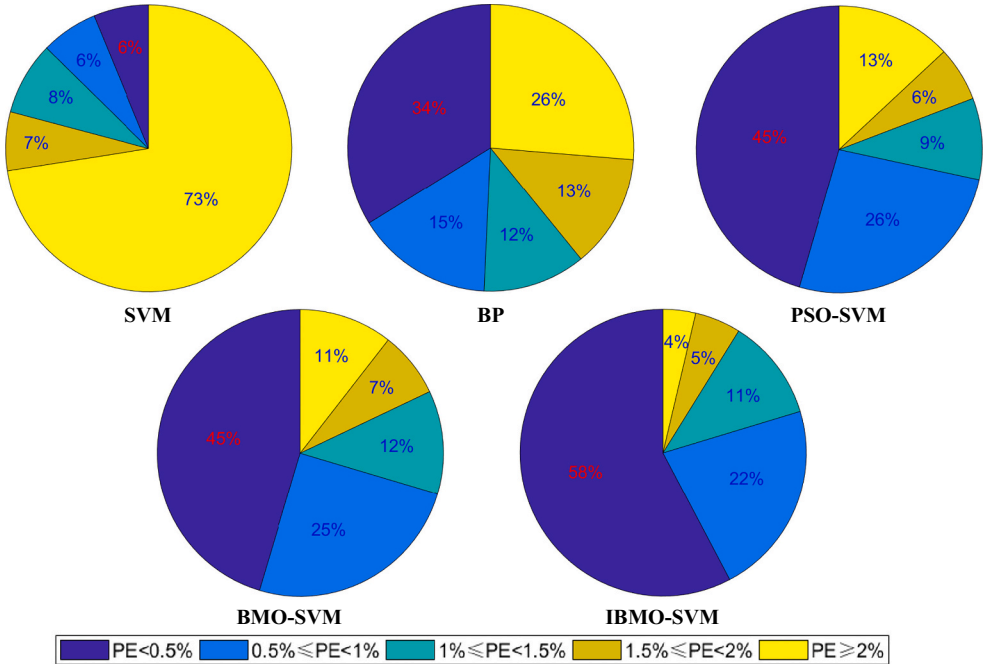


Fig. 10. Sectoral plot of model estimation error distribution.

Table 6
Error distribution of model estimation results.

Model	Number				
	$PE <$ 0.5 %	0.5 % $\leq PE <$ 1 %	1 % $\leq PE <$ 1.5 %	1.5 % $\leq PE <$ 2 %	$PE \geq 2$ %
SVM	1026	1054	1349	1086	11,910
BP	5551	2543	1905	2105	4321
PSO-SVM	7471	4292	1509	1007	2146
BMO-SVM	7452	4111	1915	1216	1731
IBMO-SVM	9480	3606	1884	850	605

Table 7
Evaluation results of each model.

Model	Evaluation index				
	RMSE	R^2	*MAPE*	*CE*	Running time (s)
SVM	0.0409	0.9451	7.64 %	545.0530	23
BP	0.0145	0.9957	1.45 %	131.1073	142
PSO-SVM	0.0072	0.9983	1.01 %	78.8423	1769
BMO-SVM	0.0063	0.9987	0.87 %	71.8181	1829
IBMO-SVM	0.0042	0.9994	0.61 %	50.7099	1843

5. Concluding remarks

This study proposes the IBMO-SVM model for lithium-ion battery SOC estimation, with the following concluding remarks:

- IBMO is proposed, which shows stronger convergence accuracy and speed than BMO for numerous test functions.
- The IBMO-SVM model is proposed and compared with four other SOC estimation models, and the comparison findings demonstrate that IBMO-SVM has the best accuracy and stability.
- The *RMSE* and *MAPE* of IBMO-SVM are only 0.0042 and 0.61 %, and R^2 is 0.9994 which is closest to 1, proving the validity and reliability of the model.
- The proposed SOC estimation model does not depend on the lithium-ion battery equivalent model, and can theoretically be applied to any chemical type of battery as long as the data used to train the model are accurate and reliable.

There are still some limitations although the proposed model has a high estimation accuracy. First, the proposed IBMO is still unable to converge all test functions to the global optimum. Second, only current, voltage, and temperature are used as influencing factors in this study, without considering other possible external factors. Finally, the battery data gathering lag is not considered in this study. The time delay of battery data collecting causes the time delay of estimation results although the model itself does not have this issue. Future research needs to focus on addressing the above limitations.

CRediT authorship contribution statement

Boying Liu: Writing – original draft, Methodology, Writing – review & editing. **Haiyu Wang:** Writing – original draft, Methodology, Writing – review & editing. **Ming-Lang Tseng:** Writing – original draft, Writing – review & editing. **Zhongtao Li:** Writing – original draft, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

Data availability

Data will be made available on request.

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